

FAHAD ALMSNED, M.D., PH.D., M.A.S.

Physician-Scientist | Expert in Bioinformatics, Clinical & Public Health Informatics | Advancing Precision & Computational Medicine | Leading Population Health Management, Digital Transformation, & Data Governance.

+966546662739 • almsned.fahad@gmail.com • https://www.linkedin.com/in/fahad-almsned/ • Al Khobar, Saudi Arabia

Summary

Accomplished physician-scientist with 16+ years in leadership, specializing in bioinformatics and precision medicine. Demonstrated success in spearheading major digital transformations, integrating AI and analytics into clinical research, and forging strategic partnerships. Expert in population health management, data governance, and regulatory compliance. Committed to data-driven approaches for enhancing patient outcomes and operational efficiency. Known for visionary leadership in personalized medicine and fostering interdisciplinary collaborations.

KEY ACHIEVEMENTS

Agile Innovation in Computational Neuroscience and Bioinformatics

Led analytical efforts with renowned NIH scientists on high-impact, agile, and streamlined omics and non-omics neuroscience projects, managing 5-10 monthly initiatives that ranged from Whole Genome Sequencing to advanced CRISPR sgRNA (single guide RNA) techniques.

Oversight of Health Data Science and Population Health Solutions

Successfully implemented 4 population segmentation and risk stratification frameworks, collaborated with the CNHI, and delivered 32 analytics projects utilizing advanced biostatistics and machine learning, driving strategic insights and innovation in healthcare at the E1 level.

Transformative Leadership in IRB Compliance and AI-Driven Protocols

Directed IRB compliance, managed 2 major inspections (conducted by NCBE and SFDA), earned an exemplary site designation from NCBE, improved operational workflows by 12% for Full Board reviews and 26% for Expedited reviews, and led the implementation of eIRB and CTMS.

Strategic Direction in R&D

Led major partnerships with top institutions, including KFUPM, Waqaya, and GMU, managing 15 research projects in precision and translational medicine. This resulted in 21 peer-reviewed publications and significantly enhanced institutional research capacity.

Pioneering Precision Medicine in SA

Co-Leading the launch of Saudi Arabia's first regional pharmacogenomics lab. Co-integrated six precision medicine pipelines with Lenovo and Oxford Nanopore Technologies, ensuring CAP accreditation. Expanded a startup's portfolio by partnering with five labs and adding 42 innovative products, significantly enhancing patient outcomes and driving growth.

PROFESSIONAL EXPERIENCE

Eastern Health Cluster (E1)

Dammam, Saudi Arabia

Group Manager, Research and Development - Academic Training Affairs and Research 12/2024 - Present

- Oversee and coordinate research and development initiatives, ensuring alignment with the strategic goals of the Academic Training Affairs and Research Department - Research Program. (Recently appointed; metrics are being established.)

King Fahad Specialist Hospital - Dammam

Dammam, Saudi Arabia

Institutional Review Board (IRB) chairperson 02/2024 - Present

- Directed the IRB team to excel in regulatory compliance, successfully navigating two major inspections conducted by the National Committee of Bioethics (NCBE) and the Saudi Food and Drug Authority (SFDA), recognizing the site as exemplary by NCBE.
- Enhanced operational workflows, boosting review efficiency with a 12% increase in Full Board Reviews and a 26% rise in Expedited Reviews.
- Led the strategic integration of AI-based medical tools within IRB processes, approving 12 innovative protocols.
- Initiated and drove the implementation of Electronic IRB (eIRB) and Clinical Trial Management (CTMS) systems, significantly advancing data management and operational efficiency, achieving 70% and 40% project completion through effective partnerships with academia.

PROFESSIONAL EXPERIENCE

King Fahad Specialist Hospital - Dammam
Dammam, Saudi Arabia

Director of Research Operations 06/2022 - Present

- Led strategic collaborations in scientific research and digital transformation, elevating the institution's profile through partnerships with King Fahd University of Petroleum and Minerals (KFUPM), the Public Health Authority (Waqaya), George Mason University (GMU), and various Centers of Excellence.
- Led and co-led diverse scientific and technical teams executing multiple intramural and extramural projects focused on translational medicine, precision medicine, oncology, genetics, and bioinformatics. Successfully managed 15 projects and contributed to 21 publications.
- Advanced the analytical functions of the Oncology Center Cancer Registry, which includes a database of over 22,000 patients. Co-authored two major strategic oncology trend reports, providing actionable insights into cancer treatment and management strategies.
- Championed continuous medical professional education, co-mentored the Mawhiba gifted program, and the newly established clinical investigator program from the Saudi Commission for Health Specialties (SCFHS).

King Fahad Specialist Hospital - Dammam
Dammam, Saudi Arabia

Associate Consultant - Public Health 06/2022 - Present

GeneoClinic
Dammam, Saudi Arabia

Chief R&D Consultant and Head of Genetic Analysis Dep. 08/2023 - Present

- Actively engaged in drafting and negotiating with CAP-accredited regional laboratories to enhance our diagnostic and prognostic product offerings. Successfully expanded our portfolio by partnering with 5 labs, adding 42 innovative products that drive improved patient outcomes and business growth.
- Co-directing the cross-departmental efforts to establish the first regional pharmacogenomics laboratory tailored to the Saudi population. This pioneering project is currently 30% complete and represents a significant stride towards personalized medicine in the region.
- Orchestrating developing and successfully launching four comprehensive precision omics-based diagnostic pipelines: Germline, Somatic, Non-Invasive Prenatal Testing, and Health and Wellness initiatives.

GeneoClinic
Dammam, Saudi Arabia

Member of the Board of Advisors 08/2023 - Present

- As a Board of Advisors member, I guide strategic direction in precision medicine and bioinformatics, establish robust data and privacy policies, and leverage my network to secure partnerships and funding.

George Mason University
Fairfax, Virginia, United States

Affiliated Graduate Faculty 08/2023 - Present

- Co-mentoring Ph.D. research project leveraging advanced machine learning algorithms to predict progression, diagnosis, and treatment responses of neurological disorders, utilizing an integrated approach with multimodal data including clinical, imaging, genetic, and omics sources.

Eastern Health Cluster (E1)
Dammam, Saudi Arabia

Exe. Director, Advanced Analytics and Predictive Modelling 08/2022 - 11/2024

- Successfully implemented and executed four comprehensive frameworks of the Bridges to Health population segmentation (Saudi Version) and risk stratification models in collaboration with the Center for National Health Insurance (CNHI). This initiative resulted in four strategic reports that were instrumental in guiding organizational strategy and were directly communicated to the E1 CEO and VP of Health Care Delivery.
- Spearheaded the development of the analytical infrastructure for the EHC-PHM unit, working in close collaboration with PricewaterhouseCoopers (PWC). This critical infrastructure supports enhanced predictive and prescriptive analytics, impacting patient care and operational efficiency.
- Directed the analysis efforts across the cluster, delivering 32 projects involving advanced biostatistics, forecasting, and machine learning. These projects have provided vital insights and strategic guidance to various executive directorates within E1, demonstrating my ability to drive significant organizational change and innovation in healthcare technology.

Novo Genomics
Riyadh (Remote), Saudi Arabia

Bioinformatics Consultant and Co-Director of the R&D Dep. 06/2020 - 06/2023

- Successfully implemented six precision medicine genomics computational pipelines (WGS, WES, NIPT, nanopore long reads) on on-premise HPC in collaboration with Lenovo and Oxford Nanopore Technologies, ensuring compliance with CAP accreditation requirements.
- Initiated NOVOVAR, a genomic tertiary analysis platform for the Saudi population, with the project 20% completed.
- Represented Novo at local and international conferences and events, delivering three talks and publishing four papers.
- Designed and established the infrastructure for the dry lab unit (100% completed) and initiated a localized LIMS system (40% completed).

PROFESSIONAL EXPERIENCE

National Institute of Neurological Disorders and Stroke (NINDS) / National Institute of Health (NIH)
Bethesda, Maryland, United States

Bioinformatics Scientist
03/2020 - 07/2022

- As a Staff Scientist, I served as the resident expert in bioinformatics and computational biology. I was Responsible for initiating and managing related projects, primarily omics-related. On average, I handle 5-10 projects per month. I meet with the requesting staff for each project to discuss it in detail and understand its needs. After that, a list of requirements, including desired deliverables, is mutually agreed upon, and a tentative action plan is outlined and executed.
- These projects include, but are not limited to, the following technologies: Whole Genome Sequencing (WGS), Whole Exome Sequencing (WES), RNA Sequencing (RNA-Seq), single-cell RNA Sequencing (scRNA-Seq), Phip-Seq, Transposable-Seq, T-Cell Receptor Sequencing (TCR-Seq), Chromatin Immunoprecipitation Sequencing (ChIP-Seq), Whole Exome Sequencing (WXS), TRAP-Seq, Chromosome Conformation Capture (Hi-C), CRISPR sgRNA, Targeted Re-Sequencing, Mass Spectrometry, Microarrays (DNA, RNA, Methylation), Quantitative Reverse Transcription PCR (Q-RT-PCR), Serology, Gene Prediction, RNA Structure Analysis, Target Prediction, Protein Structure Modeling, Promoter Modeling, Molecular Dynamics, Comparative Genomics, Phylogenetics, Systems Biology (Regulatory Networks), and ad-hoc statistics-related tasks (e.g., linear, non-linear, contingency analyses), along with Machine Learning applications. These encompass more than 35 types of different omics and non-omics data and more than 50 different tools.

Researcher Ph.D. Candidate
Fairfax, Virginia, United States

George Mason University
01/2016 - 12/2020

- I worked on my thesis projects in Prof. Jafri's lab, analyzing the molecular interactions underlying Multiple Sclerosis pathogenesis using various next-generation sequencing techniques. My skills include coding in R, Python, and BASH; differential gene expression analysis via RNA-Seq; unsupervised learning techniques; gene ontology and pathway analysis; gene regulatory network construction using PANDA-R and Cytoscape; variant calling with GATK; eQTL analysis; and machine learning. I also developed expertise in molecular dynamics simulations of proteins using NAMD.

Research Assistance
Fairfax, Virginia, United States

George Mason University
02/2018 - 02/2019

- I collaborated with Dr. Amarda and Prof. Blaisten-Barojas on high-fidelity structural modeling of the multi-domain Rift Valley fever virus L-protein, aiding the design of novel L-protein/PP1α inhibitors to treat RVFV infection, and presented this work at the 9th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics. My computational skills developed during this period include protein homology, fold recognition, ab initio modeling, MOTIF analysis, Multi-Domain Assembly, and protein structure editing and visualization using Chimera. I also gained expertise in rigid and semi-flexible protein-protein docking, molecular dynamics simulations using NAMD, and MD trajectory analyses with VMD and R packages.

King Fahad Specialist Hospital - Dammam
Dammam, Saudi Arabia

Psychiatry Resident
09/2013 - 08/2015

King Khalid Medical City Centre of Health Research
Dammam, Saudi Arabia

Head, Research Clinic
09/2013 - 08/2015

UC San Diego Health System
San Diego, California, United States

Psychiatry Clinical Trainee and Clinical Research Assistant
01/2012 - 07/2013

Weiss Memorial Hospital
Chicago, Illinois, United States

Psychiatry Clinical Trainee
08/2011 - 10/2011

King Fahad Specialist Hospital - Dammam
Dammam, Saudi Arabia

Psychiatry Resident
09/2008 - 10/2009

University of Dammam
Dammam, Saudi Arabia

Psychiatry Resident
11/2007 - 09/2008

Education

George Mason University
Fairfax, Virginia, United States

Doctor of Philosophy (Ph.D.), Bioinformatics and Computational Biology
2015 - 2020

- Phi Kappa Phi (George Mason University Chapter)

University of California, San Diego
San Diego, California, United States

Master of Advanced Studies (M.A.S.), Clinical Research
2012 - 2013

Education

Educational Commission for Foreign Medical Graduates — Chicago, Illinois, United States
ECFMG Certification — 2011 - 2011

King Faisal University — Dammam, Saudi Arabia
Bachelor of Medicine, Bachelor of Surgery (M.B.B.S.) — 1999 - 2007

Certification

Digital Leadership — LinkedIn

Leading with Innovation — LinkedIn

Master Digital Transformation — LinkedIn

Strategic Agility — LinkedIn

Accelerating Clinical Transformation for Value in Health — Oxford Value & Stewardship Programme (OVSP)

Change Management — PricewaterhouseCoopers (PWC)

Population Health Management Strategy Design — PricewaterhouseCoopers (PWC)

National Segmentation Strategy — Council Of Health Insurance (CHI)

Public Policy Analysis for Scientists and Health Professionals — Science and Technology Policy Academy

Agile Project Management Professional Certificate — Atlassian

Cloud Data Management 2023 Certified Foundations Associate — Oracle

Cloud Infrastructure 2023 Certified Foundations Associate — Oracle

Bioinformatics Single Cell Sequencing — The National Institutes of Health

Pharmacometrician Tools for Dose-Response Analysis using R — Foundation for Advanced Education in the Sciences (FAES) / NIH

Selected Publications

Identification of UBA7 expression downregulation in myelodysplastic neoplasm with SF3B1 mutations
Scientific Reports - Nature

Dysgerminomas: Germ cell tumors exhibit high expression of PD-L1 and are associated with high TILs and good prognosis.
Scientific Reports - Nature

The perception of genetic diseases and premarital screening tests in the central region of Saudi Arabia
BMC Public Health

Prospect of genetic disorders in Saudi Arabia
Frontiers in Genetics

The repertoire of CSF anti-viral antibodies in patients with neuroinflammatory diseases
Science Advances - Science

Molecular Dynamics Simulations to Decipher the Role of Phosphorylation of SARS-CoV-2 Nonstructural Proteins (NSPS) in Viral Replication
MSPI - Viruses

Differential expression of an endogenous retroviral element [HERV-K(HML-6)] is associated with reduced survival in glioblastoma patients
Scientific Reports - Nature

CSF Antibody Profiling of Patients with Multiple Sclerosis using VirScan
Multiple Sclerosis Journal - SAGE

Selected Publications

Active site prediction of phosphorylated SARS-CoV-2 N-Protein using molecular simulation
Informatics in Medicine Unlocked

In Silico Prediction of the Phosphorylation of NS3 as an Essential Mechanism for Dengue Virus Replication and the Antiviral Activity of Quercetin
MDPI - Biology

CSF antibody profiling of patients with HTLV-1 associated myelopathy/ tropical spastic paraparesis using VirScan
17th International Symposium on NeuroVirology - International Society of NeuroVirology

Transcriptomic Analysis of Multiple Sclerosis Patient-derived Monocytes By RNA-Sequencing for Candidate Gene Discovery
Informatics in Medicine Unlocked

Modeling the Tertiary Structure of the Rift Valley Fever Virus L Protein
MDPI - Molecules

Modeling the tertiary structure of a multi-domain protein: structure prediction of multi-domain proteins
The 2018 ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics

Demographic and histopathological patterns of neuro-epithelial brain tumors in Eastern Province of Saudi Arabia
Neurosciences

Expression of cytomegalovirus in glioblastoma multiforme: Myth or reality?
British Journal of Neurosurgery

Volunteering

Artificial Intelligence Governance Association
Lead, Health Intelligence Society — 04/2024 - Present

The Health Intelligence Community is a collaborative network of experts focused on advancing data-driven healthcare. We bring together researchers, clinicians, data scientists, and ethics professionals to explore innovative health applications that improve care and outcomes.

Saudi Arabian Cultural Mission
Student Mentor — 01/2011 - Present

Provide advice and guidance for newly arriving Saudi doctors in the US. The new doctors will contact by email or phone to get tips about their plans regarding USMLE exams, research and clinical training, as well as the residency application process.

MIT Hacking Medicine
Mentor — 08/2019

Helping participants develop their clinical AI projects during the hackathon.

Various Venues
Speaker

- The National Research Ethics Workshop 2024 (MoPH - Qatar)
- Participation in the Neuroethics course (GMU)
- Participation in the Systems Biology course (GMU)
- Summer Training Program (Biostatistics and Epidemiology) (KFSH-D)
- Research Methodology Workshop (KFSH-D)
- The Comprehensive Course for Molecular Genetics and Precision Medicine (KFSH-D)
- The Summer Seminar Series: Learning Neuroscience (NINDS/NIH)
- Data Driven Medicine (VMA-IAU)
- Empowering Public Health: AI-Driven Innovations in Data Collection (VMA)
- Modeling the tertiary structure of a multi-domain protein: structure prediction of multi-domain proteins (IEEE/ACM)

Global Health Exhibition
Speaker — 10/2022

Session 2: Digital transformation in healthcare: Data-driven medicine: Advancing medicine using AI and Big data

Volunteering

Reviewer

Association for Computing Machinery — 06/2019 - Present

IEEE/ACM TRANSACTIONS ON COMPUTATIONAL BIOLOGY AND BIOINFORMATICS

Mawhiba

Principal Investigator — 08/2023

Honors & awards

George Mason University Fifty-Fourth Commencement Extraordinary Student

Associated with National Institute of Neurological Disorders and Stroke (NINDS)

2019 Dissertation Completion Grant

Associated with George Mason University (GMU)

2018 SSB Bioinformatics Summer Fellowship

Associated with George Mason University (GMU)